



Unit 3 · Outcome 2 · Sustainable Development

Sustainability principles, applied to real management

KK03

VCE Environmental Science · Units 3 & 4 · Exam study summary

Key knowledge. · dot point 3 of 8 00 What this dot point actually asks The study design lists this dot point as: sustainability principles as they apply to environmental management — conservation of biodiversity and ecological integrity; efficiency of resource use; intergenerational equity; intragenerational equity; the precautionary principle; and the user-pays principle

Digital Vector EnviroSci · single-topic mastery summary with worked good-vs-bad answers, exam traps, command words and practice.



00What this dot point actually asks

Key knowledge · dot point 3 of 8 00 What this dot point actually asks The study design lists this dot point as: sustainability principles as they apply to environmental management — conservation of biodiversity and ecological integrity; efficiency of resource use; intergenerational equity; intragenerational equity; the precautionary principle; and the user-pays principle. Notice the three words that decide your marks: as they apply . VCAA isn't asking you to recite six definitions — a glossary does that. They want you to take a real management decision a

01The running case study — Latrobe Valley after coal

01 The running case study — Latrobe Valley after coal To keep every principle concrete, this whole chapter uses one decision you can picture: Victoria is shutting its brown-coal power stations, and now has to decide how to manage the land, the workers and the emissions left behind. The Latrobe Valley , about 135 km east of Melbourne, sat on one of the world's largest brown-coal (lignite) deposits and powered Victoria for a century. Hazelwood Power Station — eight 200 MW units, up to a quarter of Victoria's base-load electricity — was rated the least carb

02The six principles — definition and application

02 The six principles — definition and application Learn each one as a pair: the VCAA definition (the wording examiners reward) and the Latrobe Valley application (the marks beyond the first). The card colours match the 3D lens in the next section. P1 Biodiversity & ecological integrity Maintaining the variety of life and the health and functioning of ecosystems, so natural processes and services continue. Latrobe Valley: rehabilitating the void to a self-sustaining landform, and cutting 16 Mt CO₂/yr that drives ecosystem-damaging climate change. But flo

03Interactive · the principle lens

P·E·L WRITING

03 Interactive · the principle lens Each principle is a different lens on the same decision. Click a lens to see how that principle "reads" the Latrobe Valley transition — and where it conflicts with the others. Six lenses on one decision — click a panel in the ring or the chips below Pick a principle to view the transition through its lens. Analogy · six referees, one match Imagine the same football match watched by six referees, each only allowed to enforce one rule. The "efficiency" ref loves the renewables plan; the "intragenerational equity" ref blo



The P·E·L answer structure — make a Point, support with Evidence, then Link to the question.

04Interactive · the equity trap

⚠ EXAM TRAPS

04 Interactive · the equity trap The single most common mistake on this dot point is swapping inter generational and intra generational equity. They sound identical and they're worth different marks. This scene separates them on two axes. Toggle the two kinds of fairness: Inter — across time Intra — across now The memory hook that ends the confusion INTER = bet ween generations → fairness along the time axis (us → 2050 → 2100). INTRA = inside today's generation → fairness across people and places right now (Morwell vs Melbourne vs developing nations). In

05Applying the principles to management

05 Applying the principles to management Real management can't maximise all six at once — they pull against each other. "Applying" the principles means naming the trade-offs and judging the balance , which is exactly what "analyse" and "evaluate" questions reward. Three tensions run through the Latrobe Valley decision: Tension 1 · ecological integrity vs intragenerational equity Closing coal fast is best for the climate and the void, but it can strand Morwell's workers. Managing well means pairing the environmental goal with the Latrobe Valley Authority'

06Practise · spot the principle

06 Practise · spot the principle Each card describes a real management action. Decide which principle it most clearly serves. Instant feedback explains the trap. Next ›

07Command-word decoder

◆ COMMAND WORDS

07 Command-word decoder The verb in a question sets the ceiling on your marks. Tap each to see what the examiner is really asking for on this dot point.

08Exam traps to avoid

⚠ EXAM TRAPS

08 Exam traps to avoid Swapping inter- and intra-generational equity The number-one error. Fix: inter = across time (future generations); intra = across people now (same generation, including poorer nations). Thinking the precautionary principle means "wait until we're sure" It means the opposite. Fix: uncertainty is not a reason to delay action against serious or irreversible harm — act early. Reducing user-pays to "make customers pay more" It's about who bears the environmental cost. Fix: the polluter/user pays for repairing impacts (e.g. rehab

09P·E·L — a sentence shape for full marks

P·E·L WRITING

09 P·E·L — a sentence shape for full marks For any "apply / analyse / evaluate the principles" question, each principle you discuss should move through three beats. One principle, three sentences, three marks. P Point — name the principle and your stance State which principle and whether the management upholds or compromises it. "The user-pays principle is only partly upheld by the rehabilitation bond." E Evidence — a specific case-study fact or figure Quote a number, body or action from the case study. This is where most marks are won or lost. "The bond



The P·E·L answer structure — make a Point, support with Evidence, then Link to the question.

10Same question, two answers

✗ WEAK ANSWER

Weak response 1 / 4 Replacing coal with renewables is good for sustainability. It helps efficiency of resource use because renewables are more efficient. It is also fair for everyone and good for the environment, so it follows the sustainability principles and should be done.

✓ STRONG ANSWER & MARKING

Why it loses marks: only one principle is genuinely identified; "more efficient" and "f



The P·E·L answer structure — make a Point, support with Evidence, then Link to the question.

11Exam-style questions

★ PRACTICE & SELF-MARK

11 Exam-style questions Five graded questions, 19 marks total. Attempt each in your logbook first, then reveal the model answer, marking guide, and self-mark. Your score follows you in the dock.

★ Good vs bad worked answers

Study the contrast — the same question answered two ways. The strong column shows the moves that earn the marks.



The P-E-L answer structure — make a Point, support with Evidence, then Link to the question.

✗ WEAK ANSWER

Replacing coal with renewables is good for sustainability. It helps efficiency of resource use because renewables are more efficient. It is also fair for everyone and good for the environment, so it follows the sustainability principles and should be done. Why it loses marks: only one principle is genuinely identified; "more efficient" and "fair for everyone" are asserted with no case-study evidence and no analysis. "Good for the environment" is vague. This is a list, not an analysis.

✓ STRONG / MODEL ANSWER

Efficiency of resource use is improved: Hazelwood was the OECD's least carbon-efficient station at $\sim 1.56 \text{ t CO}_2/\text{MWh}$, so replacing it with wind and solar delivers more useful energy per unit of emissions and water. Intergenerational equity is also supported: cutting roughly 16 Mt CO_2 a year reduces the climate burden passed to future generations, helping them meet their own needs — the core of that principle. Why it earns full marks: two principles correctly named and defined, each tied to a specific figure from the case study, each linked to why it advances sustainability. Two principles $\times$ (point + applied evidence) = 4.

Digital Vector · VCE Environmental Science study summary · Sustainable Development · KK03 · digitalvector.com.au

Worked good/weak answers illustrate exam technique — always check against current VCAA marking advice.